

MELUN VILLAROCHE AP, France

WMO: 071530

Lat: 48.6106N

Lon: 2.6794E

Elev: 302

StdP: 14.54

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.7	26.1	16.5	12.8	25.6	20.7	15.7	28.0	30.7	51.1	27.7	49.9	5.6	0	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	88.2	68.9	84.2	67.5	80.8	66.1	70.9	84.1	69.0	80.9	67.4	78.0	8.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.6	99.0	75.7	64.9	93.1	73.3	63.4	88.4	71.9	34.8	84.2	33.3	81.0	32.0	78.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.3	21.0	18.5	DB	17.8	95.4	5.8	4.3	13.6	98.6	10.2	101.1	7.0	103.6	2.8	106.7
			WB	17.1	74.5	5.7	1.7	13.0	75.7	9.6	76.7	6.4	77.7	2.2	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	39.2	40.7	45.9	50.9	57.4	63.4	67.1	66.5	60.2	53.9	45.3	40.4	(6)
	DBStd	11.67	7.40	6.66	5.97	6.16	5.91	5.94	5.69	5.63	5.20	6.24	6.44	6.96	(7)
	HDD50	1327	337	263	154	60	8	0	0	0	1	32	168	304	(8)
	HDD65	4788	799	680	592	423	246	100	42	47	160	345	592	762	(9)
	CDD50	2293	3	3	27	88	237	402	530	510	307	154	26	6	(10)
	CDD65	280	0	0	0	1	10	51	107	92	17	2	0	0	(11)
	CDH74	2932	0	0	0	27	127	537	1110	948	168	15	0	0	(12)
	CDH80	929	0	0	0	1	16	155	377	348	32	0	0	0	(13)
(14) Wind	WSAvg	8.5	9.9	9.8	9.4	8.6	7.9	7.4	7.6	7.4	7.4	8.2	8.8	9.9	(14)
(15) Precipitation	PrecAvg	26.1	2.0	1.9	1.7	2.0	2.6	2.3	2.4	2.2	1.9	2.4	2.2	2.4	(15)
	PrecMax	35.6	4.5	3.6	4.5	4.6	6.2	4.8	6.0	4.1	3.9	5.2	4.3	5.9	(16)
	PrecMin	20.0	0.2	0.3	0.3	0.2	0.2	0.3	0.8	0.3	0.0	0.8	0.6	0.7	(17)
	PrecStd	3.9	0.9	0.9	1.0	1.2	1.2	1.1	1.3	1.1	1.1	1.0	1.0	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	60.5	68.6	77.6	82.2	90.3	93.9	95.4	84.2	76.0	63.6	57.1	(19)
		MCWB	53.4	51.4	54.5	62.0	67.3	71.3	70.4	71.1	65.9	63.5	58.2	54.4	(20)
	2%	DB	54.0	54.9	63.6	72.4	77.6	84.3	87.9	88.3	79.0	70.6	60.0	54.9	(21)
		MCWB	51.5	49.6	53.4	58.1	63.8	68.8	68.9	68.3	64.7	61.9	56.3	52.6	(22)
	5%	DB	51.7	52.4	59.3	67.2	73.5	80.0	84.0	83.6	74.4	66.4	57.3	52.9	(23)
		MCWB	49.3	49.0	51.7	55.7	61.8	67.0	67.4	67.0	62.9	59.9	54.3	50.9	(24)
	10%	DB	49.3	50.3	55.9	62.8	69.5	76.0	80.4	79.2	70.9	63.2	54.7	50.7	(25)
		MCWB	47.2	47.5	50.2	53.5	59.8	64.6	66.0	65.2	61.2	58.2	52.3	48.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.3	53.2	56.8	62.9	69.1	73.6	72.7	72.6	68.4	65.4	59.4	55.3	(27)
		MCDB	55.9	56.7	64.3	75.0	79.8	86.6	88.2	88.7	79.2	72.4	61.9	56.6	(28)
	2%	WB	51.9	50.9	54.8	59.4	65.5	70.4	70.4	70.4	66.1	62.8	56.8	53.0	(29)
		MCDB	53.7	53.6	60.8	69.7	75.1	81.1	84.2	83.9	75.3	68.6	59.7	54.5	(30)
	5%	WB	49.5	49.4	52.8	56.7	62.9	67.9	68.6	68.3	64.3	60.9	54.7	51.0	(31)
		MCDB	51.4	51.9	57.8	65.1	70.9	77.9	81.0	80.0	72.3	65.5	57.1	52.7	(32)
	10%	WB	47.4	47.6	50.9	54.4	60.6	65.6	67.0	66.5	62.5	59.0	52.6	48.9	(33)
		MCDB	49.1	50.2	55.4	61.4	68.2	74.2	77.8	76.4	69.2	62.5	54.6	50.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	12.1	15.9	18.7	18.3	19.5	20.3	20.5	18.8	15.2	10.8	9.2	(35)
		MCDBR	11.0	15.1	22.1	25.8	25.1	26.2	27.7	28.5	24.9	19.2	13.3	10.4	(36)
		MCWBR	9.3	11.0	13.4	13.8	13.1	13.1	11.0	11.6	12.3	11.4	9.9	8.8	(37)
	5% WB	MCDBR	10.6	13.7	18.3	22.9	22.7	24.1	23.9	24.7	20.9	17.2	12.2	10.1	(38)
		MCWBR	9.4	10.7	12.1	13.0	12.8	13.1	10.6	11.3	11.4	10.9	9.7	9.1	(39)
(40) Clear-Sky Solar Irradiance	taub		0.331	0.352	0.386	0.411	0.417	0.421	0.420	0.409	0.379	0.371	0.343	0.327	(40)
	taud		2.398	2.349	2.251	2.196	2.220	2.245	2.244	2.297	2.363	2.376	2.403	2.371	(41)
	Ebn at Noon		230	251	260	265	268	267	266	263	260	240	222	213	(42)
	Edh at Noon		22	28	36	42	43	42	42	38	32	27	21	20	(43)
(44) All-Sky Solar Radiation	RadAvg		290	542	923	1388	1640	1831	1730	1510	1160	691	351	243	(44)
	RadStd		35	78	113	156	164	166	146	148	103	53	34	41	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page